

Gerindra Adi

Amherst, MA | 646-267-0184 | gerindraadi@gmail.com | [linkedin.com/in/gerindra-adi](https://www.linkedin.com/in/gerindra-adi) | geribydesign.com

EDUCATION

University of Massachusetts Amherst

Manning College of Computer Sciences

B.S. in Computer Science

Exp. Graduation: May 2026

Amherst, MA

GPA: 3.98

Awards: Chancellor's Award Scholarship (\$16k/yr), Dean's List: Fall 2022 - Spring 2025

Coursework: Machine Learning (*Python, PyTorch*), Artificial Intelligence (*Python*), Scalable Web Systems (*Docker, FastAPI, REST APIs*), Data Management (*PostgreSQL*), Data Structures, Web Programming

PUBLICATION

Tsalah, J.M., **Adi, G.**, Elkinton, J.S., & Andersen, J.C. (2026). Wing morphometrics allows for rapid and accurate identifications of the invasive winter moth. *Agricultural and Forest Entomology*. (Under Review)

EXPERIENCE

Undergraduate Machine Learning Researcher

Aug. 2025 – Present

Forest Insect Conservation and Management Lab

Amherst, MA

- Co-authored manuscript submitted to *Agricultural and Forest Entomology* implementing **K-means clustering** and **SVM** classification on Procrustes-aligned morphometric data, achieving 91.2% accuracy identifying invasive winter moth across 59 vouchered specimens
- Built data pipelines to clean and process 1,000+ moth-wing specimens, performing morphometric feature extraction and Procrustes alignment for model training
- Engineering **computer vision** classification system using **YOLOv11** and **scikit-learn** for a moth-wing identification application, implementing **model training**, validation against ground-truth labels, and **inference optimization** for deployment

Undergraduate Researcher

May 2025 – Aug. 2025

Human-Computer Interaction and Visualization Lab

Amherst, MA

- Co-authored research study evaluating LLM-generated health visualizations, conducting qualitative analysis on AI model outputs across user demographics
- Built automated data generation pipeline integrating GPT-4 and Claude APIs, processing health tracker datasets and generating 300+ AI-powered visualizations for research analysis

Undergrad Course Assistant (Scalable Web Systems)

Sept. 2024 – Present

University of Massachusetts Amherst

Amherst, MA

- Held weekly **office hours** supporting 60+ students on **scalability**, **FastAPI**, **middleware**, **SDLC**, **Git**, and **documentation** best practices
- Collaborated with staff to develop materials on **CI/CD** practices, **SQL**-based **Prisma** backends, and **Amazon AWS S3** media storage for production-grade student projects

PROJECTS

Timeline | *Python, TypeScript, React, Docker, PostgreSQL*

Sept. 2025 - Present

- Co-developing containerized full-stack application using Docker for local development and deployment; implementing RESTful API architecture with PostgreSQL backend and unit testing

Songbird | *Python, Anthropic API, React*

Aug. 2025 - Present

- Built AI-powered recommendation engine integrating Anthropic's Claude API, implementing prompt engineering techniques for personalized inference-time generation

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, C, SQL, Bash, HTML, CSS

ML & AI: PyTorch, scikit-learn, YOLOv11, SVM, K-means, PCA, pandas, NumPy, Jupyter Notebooks, prompt engineering, LLM API integration (OpenAI, Anthropic)

Web & Infrastructure: React, Node.js, FastAPI, REST APIs, Prisma, PostgreSQL, Docker, AWS S3, CI/CD, Git, Linux/Unix, Agile/Scrum